

- 2 A student places a steel strip of thickness d and width w between two blocks that are fixed to the bench. A load is added to the strip as shown in Figure 2.1.

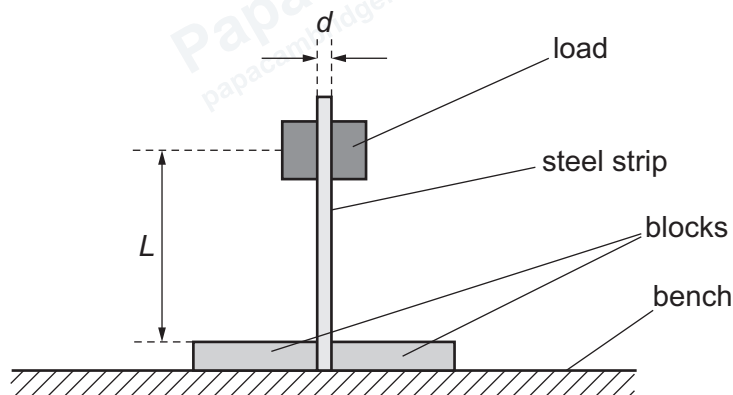


Figure 2.1

The distance between the blocks and the centre of gravity of the load is L .

The load is displaced a small distance and released so that it oscillates. The time t for 20 oscillations is measured. The period T of the oscillations is calculated.

The experiment is repeated for steel strips with the same thickness but different widths.

It is suggested that T and w are related by the equation

$$E = \frac{16\pi^2 ML^3}{wT^2 d^3}$$

where E is the Young modulus of steel and M is the mass of the load.

- (a) A graph is plotted of T^2 on the y -axis against $\frac{1}{w}$ on the x -axis.

Determine an expression for the gradient.

gradient = [1]

- (b) Values of w , $\frac{1}{w}$ and t are given in Table 2.1.

Table 2.1

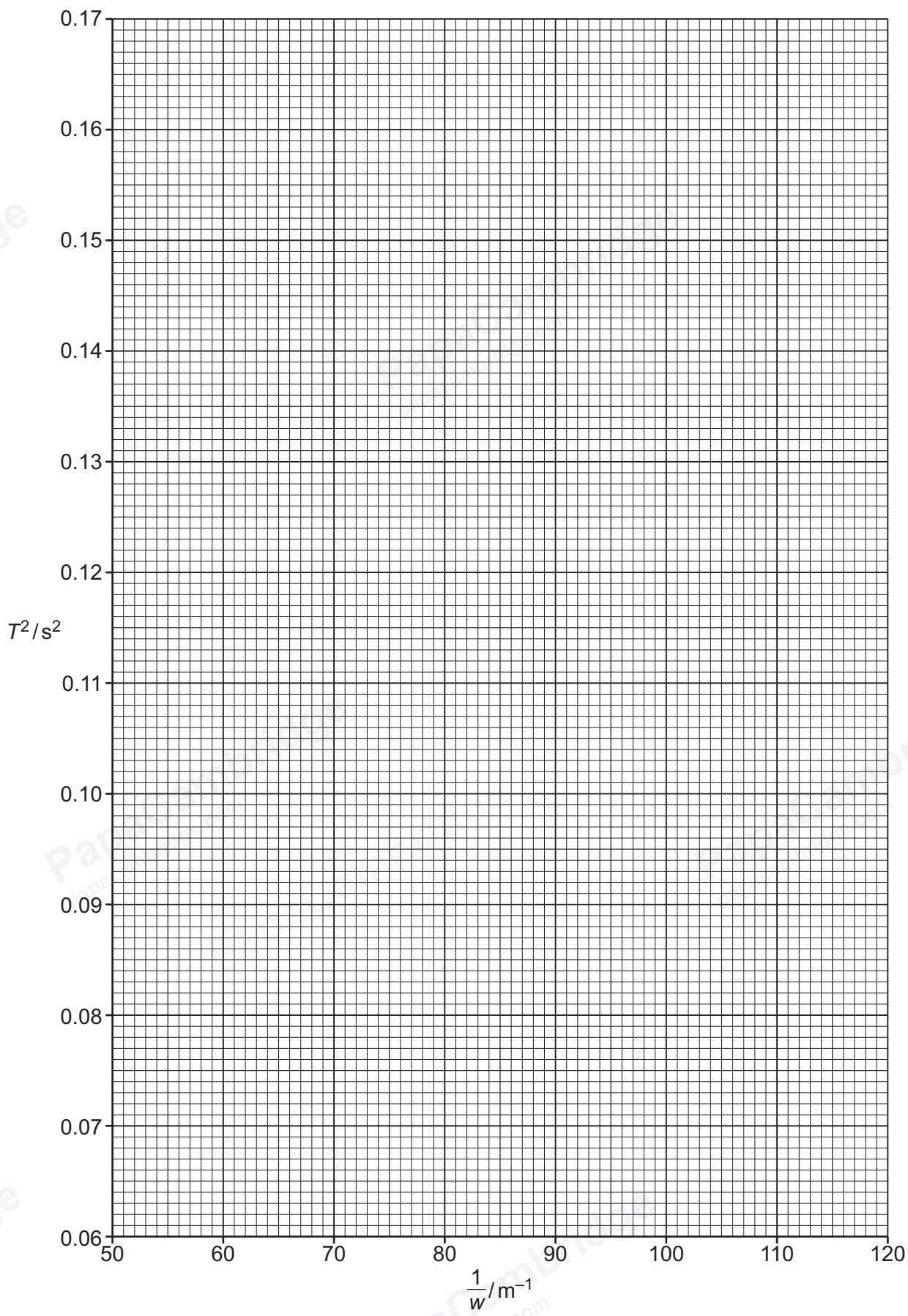
w/mm	$\frac{1}{w}/\text{m}^{-1}$	t/s	T/s	T^2/s^2
8.5	118	8.00 ± 0.08		
9.8	102	7.42 ± 0.08		
11.7	85.5	6.78 ± 0.08		
14.1	70.9	6.20 ± 0.08		
16.3	61.3	5.78 ± 0.08		
19.2	52.1	5.42 ± 0.08		

Calculate and record values of T/s and T^2/s^2 in Table 2.1. Include the absolute uncertainties in T and T^2 .

[2]

- (c) (i) Plot a graph of T^2/s^2 against $\frac{1}{w}/\text{m}^{-1}$. Include error bars for T^2 . [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]
- (iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



(d) The thickness of the strips is measured several times:

0.98 mm

0.99 mm

0.93 mm

0.97 mm

0.98 mm.

Determine the mean thickness d . Include the absolute uncertainty.

$d = \dots\dots\dots$ mm [1]

(e) (i) Using your answers to **2(a)**, **2(c)(iii)** and **2(d)**, determine the value of E . Include an appropriate unit.

Data: $M = (100 \pm 5)\text{g}$
 $L = (26.2 \pm 0.2)\text{cm}$

$E = \dots\dots\dots$ [2]

(ii) Determine the percentage uncertainty in E .

percentage uncertainty in $E = \dots\dots\dots$ % [1]

(f) The experiment is repeated. Determine the width w that gives a value of T of $(0.500 \pm 0.004)\text{s}$. Include the absolute uncertainty.

$w = \dots\dots\dots$ m [2]

[Total: 15]